

If the current estimate of the PHD is $(\mathbf{a}, \mathbf{T})$, then the conditional expectations of N_i , x_i and N_{ij} (E-step) are given by

$$E_{(\mathbf{a}, \mathbf{T})}[N_i] = \frac{1}{n} \sum_{k=1}^n \frac{\mathbf{a}_i (e^{\mathbf{T}v_k} \mathbf{T}^0)_i}{\mathbf{a} e^{\mathbf{T}v_k} \mathbf{T}^0}, \quad E_{(\mathbf{a}, \mathbf{T})}[x_i] = \frac{1}{n} \sum_{k=1}^n \frac{\left[\int_0^{v_k} (\mathbf{a} e^{\mathbf{T}(v_k-u)})' (e^{\mathbf{T}u} \mathbf{T}^0)' du \right]_{ii}}{\mathbf{a} e^{\mathbf{T}v_k} \mathbf{T}^0},$$

$$E_{(\mathbf{a}, \mathbf{T})}[N_{ij}] = \frac{1}{n} \sum_{k=1}^n \frac{\left[\int_0^{v_k} (\mathbf{a} e^{\mathbf{T}(v_k-u)})' (e^{\mathbf{T}u} \mathbf{T}^0)' du \right]_{ij}}{\mathbf{a} e^{\mathbf{T}v_k} \mathbf{T}^0} \mathbf{T}_{ij}, \quad E_{(\mathbf{a}, \mathbf{T})}[N_{i,m+1}] = \frac{1}{n} \sum_{k=1}^n \frac{(\mathbf{a} e^{\mathbf{T}v_k})_i \mathbf{T}_i^0}{\mathbf{a} e^{\mathbf{T}v_k} \mathbf{T}^0}$$

Then, the M-step results in the estimation of new parameters

$$\hat{\mathbf{a}}_i = E_{(\mathbf{a}, \mathbf{T})}[N_i]; \quad \hat{\mathbf{T}}_{ij} = \frac{E_{(\mathbf{a}, \mathbf{T})}[N_{ij}]}{E_{(\mathbf{a}, \mathbf{T})}[x_i]}, \quad i \neq j; \quad \hat{\mathbf{T}}_i^0 = \frac{E_{(\mathbf{a}, \mathbf{T})}[N_{i,m+1}]}{E_{(\mathbf{a}, \mathbf{T})}[x_i]}; \quad \hat{\mathbf{T}}_{ii} = - \left(\hat{\mathbf{T}}_i^0 + \sum_{\substack{j=1 \\ j \neq i}}^m \hat{\mathbf{T}}_{ij} \right)$$

4. RESULTS AND DISCUSSION

To analyze the behavior of V_{reset} , the classical methodology based on the Weibull distributions has been used as it can be seen in Section 2. The results are not as good as desirable. Phase-type distributions **with their corresponding** properties have been introduced in the section above. One interesting property of PHD is that this class of distributions is dense in the non-negative probability distributions set. Thus, PHD are going to be assumed to estimate V_{reset} distribution.

The voltage up to the conductive filament failure has been fitted by considering multiple general PHDs with different stages by using the EM algorithm [38] described in Subsection 3.3. The computations have been made by using the program EMpht for fitting phase-type distributions to data [40], in addition to developing own code with the software R and using the R project for Statistical Computing [41].

Thirty PHD with m transient stages, for $m = 1, \dots, 30$, have been fitted to our data set by using the *EM*-algorithm. In total $n=2749$ V_{reset} were observed. We have assumed any internal structure for matrix $\mathbf{T}$ (transition intensities), therefore for each one we have estimated $m(m-1)+m$ parameters. After this analysis, we have observed that the internal structure of the PHD representation depends only of one parameter, fixed m , for all cases. This structure can be expressed as follows

$$\boldsymbol{\alpha} = (1, 0, \dots, 0) \text{ and } \mathbf{T} = \begin{pmatrix} -\lambda & \lambda & 0 & 0 & \dots \\ 0 & -\lambda & \lambda & 0 & \dots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & -\lambda & \lambda \\ 0 & \dots & \dots & 0 & -\lambda \end{pmatrix}. \quad (5)$$

It is known that a PHD with the structure described above is a Erlang distribution with representation $E(m, \lambda)$ as it can be seen in section 3.2. The cumulative distribution function of an Erlang distribution is given by

$$F(v) = 1 - \sum_{j=0}^{m-1} e^{-\lambda v} (\lambda v)^j / j!. \quad (6)$$

Thus, for the representation described in (5), both distributions: phase-type and Erlang, are equivalent, but now Erlang distribution is expressed in an algorithmic way through PH distributions. Therefore, functions (2) and (6) are the same.

Taking into account the previous results, we can conclude from the PHD analysis that the voltage up to the rupture process (V_{reset}) is Erlang distributed; however, the PHD structure is considered. The physical structure of the Erlang distribution and its parameters can be interpreted as follows: the conductive filament always begins in stage 1 (after a successful set process where its formation has been achieved) and it undergoes a sequential degradation thorough m distinct and well differenced stages (the number of

stages is characterized by parameter m) where the mean V_{reset} in each stage is equal to $1/\lambda$ (inverse of the second parameter of the Erlang distribution).

The Erlang distribution parameters have been estimated. Table 1 shows the λ estimates after applying the EM algorithm for different stages (whose number is described by parameter m).

Number of stages	Iterations		Estimate λ
	EM algorithm	LogL	
15	1700	-1066.427	9.279325
14	1700	-1096.136	8.660703
13	1300	-1133.043	8.042082
12	1000	-1178.317	7.423459
11	1000	-1233.436	6.804838
10	800	-1300.308	6.186217
9	600	-1381.457	5.567595
8	600	-1480.314	4.948974
7	600	-1601.724	4.330352
6	500	-1752.833	3.711730
5	200	-1944.833	3.093108
4	200	-2196.729	2.474486
3	200	-2544.869	1.855864

Table 1. Parameter λ estimated by using the EM algorithm depending on the number of stages.

The optimum value is reached for 15 stages with $\hat{\lambda} = 9.279325$. Therefore, the estimated mean V_{reset} in each stage is equal to 0.1078. Finally, the mean estimated V_{reset} from the beginning up to the conductive filament failure is 1.6165.

The experimental cumulative hazard rate estimated by the Erlang and Weibull distributions have been plotted and compared graphically (see Figure 3).